Problem-Solving, Ratios, Rates & Data

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • [FRQ] • Target: Multi-Part Ratios & Proportional Partitioning**

In a concrete mix, the ratio of cement to sand to gravel by weight is $1:2.5:4$. How many kilograms of each material are needed to produce $600\,\text{kg}$ of concrete?

Total ratio parts $= 1 + 2.5 + 4 = 7.5$. Each part equals $\frac{600}{7.5} = 80\,\text{kg}$. Thus cement $= 80\,\text{kg}$, sand $= 2.5(80) = 200\,\text{kg}$, and gravel $= 4(80) = 320\,\text{kg}$.

2 **[Foundational] • [FRQ] • Target: Dimensional Analysis & Unit Conversion**

Convert a speed of $90\,\text{km/h}$ into meters per second (m/s) and feet per second (ft/s, using $1\,\text{m} \approx 3.28\,\text{ft}$).

$90\,\text{km/h} \times \frac{1000\,\text{m}}{1\,\text{km}} \times \frac{1\,\text{h}}{3600\,\text{s}} = \frac{90{,}000}{3600} = 25\,\text{m/s}$. In feet per second, $25 \times 3.28 = 82.0\,\text{ft/s}$.

3 **[Foundational] • [FRQ] • Target: Successive Percentage Changes**

An electronics store marks up the wholesale cost of a laptop by 40% to set the retail list price, and then offers a 25% holiday discount off the list price. Find the net percentage profit over wholesale cost.

Multiply the sequential scale factors: $1.40 \times (1 - 0.25) = 1.40 \times 0.75 = 1.05$. A combined multiplier of 1.05 represents a net profit of 5% above wholesale cost.

4 **[Foundational] • [FRQ] • Target: Reverse Percentage Problems**

After a 16% price reduction, a laboratory microscope sells for 630 dollars. What was the original price of the microscope before the reduction?

Let P_0 be the original price: $(1 - 0.16)P_0 = 0.84P_0 = 630 \Rightarrow P_0 = \frac{630}{0.84} = 750$ dollars.

5 **[Foundational] • [FRQ] • Target: Scale Drawings: Linear vs. Area Scaling**

On an architectural blueprint drawn at a scale of $1\,\text{cm} : 2.5\,\text{m}$, a rectangular conference room measures $4.8\,\text{cm}$ by $3.2\,\text{cm}$. Find the actual perimeter (in meters) and actual floor area (in m^2) of the room.

Actual length $= 4.8 \times 2.5 = 12\,\text{m}$; actual width $= 3.2 \times 2.5 = 8\,\text{m}$. Actual perimeter $= 2(12 + 8) = 40\,\text{m}$, and actual floor area $= 12 \times 8 = 96\,\text{m}^2$.

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6 [Foundational] • [FRQ] • Target: Combined Work Rates

Pump A can fill a reservoir in 6 hours, and Pump B can fill the same reservoir in 10 hours. If both pumps operate together, how many hours does it take to fill the reservoir?

Combined rate: $R = \frac{1}{6} + \frac{1}{10} = \frac{5+3}{30} = \frac{8}{30} = \frac{4}{15}$ reservoir per hour. Time required $= \frac{1}{R} = \frac{15}{4} = 3.75$ hours (3 hours 45 minutes).

7 [Foundational] • [FRQ] • Target: Reading Rates & Percentages from Bar Charts

The bar chart below shows quarterly solar energy output (in MWh) for a microgrid across four quarters. Calculate the percentage increase from Q1 to Q3 and the percentage of annual output generated in Q3.

From Q1 (120 MWh) to Q3 (192 MWh), the percentage increase is $\frac{192-120}{120} \times 100\% = 60\%$. Total annual output is $120 + 150 + 192 + 138 = 600$ MWh, so Q3 accounts for $\frac{192}{600} \times 100\% = 32\%$.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Concentration Enrichment Equations

A metallurgist has 80 g of a 35% silver alloy. How many grams of pure (100%) silver must be melted into the alloy to raise the silver concentration to 60%?

Let x be the grams of pure silver added: $\frac{0.35(80)+1.00x}{80+x} = 0.60 \Rightarrow 28 + x = 48 + 0.60x \Rightarrow 0.40x = 20 \Rightarrow x = 50$ g.

9 [Intermediate] • [FRQ] • Target: Inverse-Square Variation & Percentage Change

The illumination intensity I from a point light source varies inversely as the square of the distance d from the source ($I = k/d^2$). If $I = 180$ lux at $d = 2$ m, find I at $d = 6$ m, and determine by what percentage I changes when d increases by 25%.

At $d = 6$ m (3 times the distance), $I = \frac{180}{3^2} = 20$ lux. When d is multiplied by $1.25 = \frac{5}{4}$, I is multiplied by $\left(\frac{4}{5}\right)^2 = \frac{16}{25} = 0.64$, which is a 36% decrease.

10 [Intermediate] • [FRQ] • Target: Weighted Average Speed Across Segments

A commuter drives the first 60 km of a trip at 80 km/h and the remaining 90 km at 60 km/h. Compute the total travel time and the overall average speed for the 150 km trip.

Time for segment 1: $t_1 = \frac{60}{80} = 0.75$ h. Time for segment 2: $t_2 = \frac{90}{60} = 1.50$ h. Total time $= 2.25$ h $= \frac{9}{4}$ h, so overall average speed $= \frac{150}{9/4} = \frac{600}{9} = \frac{200}{3} \approx 66.67$ km/h.

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11 **[Intermediate]** • **[FRQ]** • **Target: Percentage Inclusion-Exclusion**

In a student body of $1{,}200$ students, 65% take Spanish, 45% take French, and 20% take neither language. How many students take both Spanish and French, and what percentage of Spanish students also take French?

Students taking at least one language: 80% of $1{,}200 = 960$. By Inclusion-Exclusion, $65\% + 45\% - b = 80\% \Rightarrow b = 30\%$, which is $0.30(1200) = 360$ students. Among Spanish students (65%), the fraction also taking French is $\frac{30\%}{65\%} = \frac{6}{13} \approx 46.15\%$.

12 **[Intermediate]** • **[FRQ]** • **Target: Multi-Step Rate Unit Conversion**

Convert an irrigation flow rate of 450 gallons per hour into liters per minute, given $1\,\text{gallon} = 3.785\,\text{L}$. Round to two decimal places.

$450\,\frac{\text{gal}}{\text{h}} \times \frac{3.785\,\text{L}}{1\,\text{gal}} \times \frac{1\,\text{h}}{60\,\text{min}} = 7.5 \times 3.785 = 28.3875 \approx 28.39\,\text{L/min}.$

13 **[Intermediate]** • **[FRQ]** • **Target: Compound Percentage Recovery**

A stock portfolio drops by 20% in Year 1 and drops by another 10% in Year 2. By what percentage must the portfolio grow in Year 3 to return to its exact initial value?

After two years, the value multiplier is $0.80 \times 0.90 = 0.72$. To return to 1.00, the Year 3 multiplier must be $\frac{1}{0.72} = \frac{25}{18} \approx 1.3889$, which requires a 38.89% (or $\frac{350}{9}\%$) gain.

14 **[Intermediate]** • **[FRQ]** • **Target: Chained Ratios & Homogeneous Expressions**

If $a : b = 3 : 4$ and $b : c = 6 : 5$, find the compound ratio $a : b : c$ in lowest integers and evaluate $\frac{2a + 3c}{4b}$.

Match b using $\text{lcm}(4, 6) = 12$: $a : b = 9 : 12$ and $b : c = 12 : 10$, so $a : b : c = 9 : 12 : 10$. Then $\frac{2a + 3c}{4b} = \frac{2(9) + 3(10)}{4(12)} = \frac{18 + 30}{48} = \frac{48}{48} = 1$.

Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **[FRQ]** • **Target: Invariant Dry-Matter Percentage Problems** • **Cross-Lesson: Topic 1.01 (Conservation Linear Equations)**

Fresh berries contain 80% water by weight, whereas dried berries contain 20% water by weight. How many kilograms of dried berries are obtained from $50\,\text{kg}$ of fresh berries?

The non-water dry solids remain invariant during drying: $50\,\text{kg}$ of fresh berries contain $(1 - 0.80)(50) = 10\,\text{kg}$ of dry solids. In dried berries, dry solids constitute $1 - 0.20 = 80\%$ of the total mass M: $0.80M = 10 \Rightarrow M = 12.5\,\text{kg}$.

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16 **[Challenging] • [FRQ] • Target: Power-Law Proportionality & Percentage Scaling • Cross-Lesson: Topic 4.01 (Fractional Exponents)**

According to Kepler's Third Law, the square of a planet's orbital period T is directly proportional to the cube of its semi-major axis a ($T^2 = ka^3$). If Planet X has a semi-major axis that is 4 times that of Earth ($a_E = 1\,\text{AU}, T_E = 1\,\text{yr}$), find Planet X's orbital period and the percentage increase in T when a increases by 21%.

For $a = 4$, $T = 4^{3/2} = 8$ Earth years. When a increases by 21%, the scale factor for a is $1.21 = (1.1)^2$, so T scales by $(1.21)^{3/2} = (1.1)^3 = 1.331$, which is a 33.1% increase.

17 **[Challenging] • [FRQ] • Target: Similarity Scaling of 2D Area & 3D Volume • Cross-Lesson: Topic 8.01 (3D Solids & Similarity)**

Two similar right circular cones have surface areas in the ratio $9:25$. Find (a) the ratio of their heights, (b) the ratio of their volumes, and (c) the percentage by which the larger cone's volume exceeds the smaller cone's volume.

(a) Linear dimensions scale as the square root of area ratio: $h_1 : h_2 = \sqrt{9} : \sqrt{25} = 3:5$. (b) Volumes scale as the cube of linear ratio: $V_1 : V_2 = 3^3 : 5^3 = 27:125$. (c) Percentage excess $= \frac{125-27}{27} \times 100\% = \frac{9800}{27}\% \approx 362.96\%$.

18 **[Challenging] • [FRQ] • Target: Relative Risk, Sensitivity & Bayesian Proportions • Cross-Lesson: Topic 6.01 (Conditional Probability & Two-Way Tables)**

A medical screening test for a condition present in 2% of a population has a sensitivity (true-positive rate) of 95% and a specificity (true-negative rate) of 90%. Out of $10{,}000$ people tested, how many test positive in total, and what percentage of those who test positive actually have the condition (Positive Predictive Value)?

Out of $10{,}000$: 200 have the condition ($0.95 \times 200 = 190$ true positives) and $9{,}800$ do not ($0.10 \times 9800 = 980$ false positives). Total positive tests $= 190 + 980 = 1{,}170$. Positive Predictive Value $= \frac{190}{1170} = \frac{19}{117} \approx 16.24\%$.

19 **[Challenging] • [FRQ] • Target: Repeated Percentage Decay & Exponential Inequalities • Cross-Lesson: Topic 4.01 (Exponential Decay) & Topic 2.01 (Inequalities)**

Under continuous-rate radioactive decay, a sample loses 10% of its remaining mass every hour. Write an inequality for the number of hours t until less than 50% of the initial mass remains, and determine the smallest integer hour t that satisfies it.

The remaining fraction after t hours is $(0.90)^t < 0.50$. Testing integer powers: $(0.9)^6 = 0.531441 > 0.50$ and $(0.9)^7 = 0.4782969 < 0.50$. Thus the smallest integer hour is $t = 7$ hours.

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20 [Challenging] • [FRQ] • Target: Quadratic Revenue Elasticity from Percentage Changes • Cross-Lesson: Topic 3.01 (Quadratic Functions & Roots)

An airline increases its ticket price by $p\%$ and observes that ticket sales volume decreases by $\frac{p}{2}\%$. Express the percentage change in total revenue as a quadratic function of p, and find the non-zero value of p for which total revenue remains unchanged.

Revenue multiplier is $\left(1+\frac{p}{100}\right)\left(1-\frac{p}{200}\right)=1+\frac{p}{200}-\frac{p^2}{20000}$. For revenue to remain unchanged, $\frac{p}{200}-\frac{p^2}{20000}=0\Rightarrow p(100-p)=0\Rightarrow p=100\%$ (price doubles, volume drops by 50%, revenue multiplier $=2\times0.5=1$).

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Simplifying Algebraic Ratios

If $\frac{3x}{5y}=\frac{9}{10}$, what is the ratio of x to y?

a. $2:3$

b. $3:2$ ✓

c. $9:50$

d. $27:50$

Multiply both sides by $\frac{5}{3}$: $\frac{x}{y}=\frac{9}{10}\cdot\frac{5}{3}=\frac{3}{2}$, or $3:2$.

22 [Foundational] • [MCQ] • Target: Discount & Tax Calculations

A shirt originally priced at 60 dollars is on sale for 25% off. If an 8% sales tax is applied to the sale price, what is the final amount paid?

a. 45.00 dollars

b. 48.60 **dollars** ✓

c. 49.80 dollars

d. 51.00 dollars

Sale price $=60(0.75)=45.00$ dollars. Adding 8% sales tax gives $45.00(1.08)=48.60$ dollars.

23 [Foundational] • [MCQ] • Target: Unit Rates & Time Conversion

A 3D printer produces 18 identical gears in 45 minutes. At this constant rate, how many gears can it produce in 2.5 hours?

a. 45

b. 54

c. 60 ✓

d. 75

Rate $=\frac{18\,\text{gears}}{45\,\text{min}}=0.4\,\text{gears/min}$. In $2.5\,\text{hours}=150\,\text{minutes}$, output is $0.4\times150=60$ gears.

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24 [Foundational] • [MCQ] • Target: Equating Percentage Expressions

If 40% of x is equal to 16% of 150, what is the value of x?

a. 24

b. 48

c. 60 ✓

d. 96

$0.40x = 0.16(150) = 24 \Rightarrow x = \frac{24}{0.40} = 60$.

25 [Foundational] • [MCQ] • Target: Direct Proportions & Map Scales

A map uses a scale of 1 inch $= 15$ miles. If two cities are 4.6 inches apart on the map, what is the actual straight-line distance between them?

a. 60 miles

b. 64 miles

c. 69 miles ✓

d. 75 miles

Multiply $4.6 \times 15 = 69$ miles.

26 [Foundational] • [MCQ] • Target: Percentage Change Formula

The price of a commodity increased from 80 dollars to 100 dollars. What was the percentage increase?

a. 20%

b. 25% ✓

c. 80%

d. 125%

Percentage increase $= \frac{100-80}{80} \times 100\% = \frac{20}{80} \times 100\% = 25\%$.

27 [Foundational] • [MCQ] • Target: Part-to-Whole Ratio Calculations

A fruit punch recipe calls for orange juice and pineapple juice in a ratio of $5:3$. How many liters of orange juice are in 24 liters of punch?

a. 9

b. 12

c. 15 ✓

d. 18

Orange juice makes up $\frac{5}{5+3} = \frac{5}{8}$ of the punch: $\frac{5}{8} \times 24 = 15$ liters.

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28 [Foundational] • [MCQ] • Target: Direct Variation

If y is directly proportional to x, and $y = 18$ when $x = 12$, what is the value of y when $x = 20$?

a. 24

b. 26

c. 30 ✓

d. 32

The constant of proportionality is $k = \frac{18}{12} = 1.5$. When $x = 20$, $y = 1.5(20) = 30$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 [Intermediate] • [MCQ] • Target: Compound Percentage Change in Area

The length of a rectangle is increased by 20% and its width is decreased by 15%. How does the area of the rectangle change?

a. It increases by 5%.

b. **It increases by 2%.** ✓

c. It decreases by 2%.

d. It remains unchanged.

Multiply the dimension scale factors: $(1.20)(0.85) = 1.02$, which is a 2% increase in area.

30 [Intermediate] • [MCQ] • Target: Inverse-Square Proportions

Sound intensity I is inversely proportional to the square of distance r from the speaker. If a listener moves from $r = 3\,\text{m}$ to $r = 6\,\text{m}$, the new intensity is what fraction of the original intensity?

a. $\frac{1}{16}$

b. $\frac{1}{4}$ ✓

c. $\frac{1}{3}$

d. $\frac{1}{2}$

Doubling the distance ($r \to 2r$) divides the intensity by $2^2 = 4$, giving $\frac{1}{4}$ of the original intensity.

31 [Intermediate] • [MCQ] • Target: Evaporation & Concentration Ratios

How many liters of water must be evaporated from 50 L of a 12% salt solution to produce a 20% salt solution?

a. 15 L

b. 20 L ✓

c. 25 L

d. 30 L

The amount of dissolved salt is constant at $0.12(50) = 6\,\text{L}$. For 6 L to be 20% of the final volume V, $0.20V = 6 \Rightarrow V = 30\,\text{L}$, so $50 - 30 = 20\,\text{L}$ of water must be evaporated.

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32 [Intermediate] • [MCQ] • Target: Algebraic Reverse Percentages

After a 20% discount, the sale price of a coat is p dollars. Which expression represents the original price of the coat before the discount?

a. $0.80p$

b. $1.20p$

c. $1.25p$ ✓

d. $\frac{p}{1.20}$

Since $0.80P_0 = p$, we have $P_0 = \frac{p}{0.80} = \frac{5}{4}p = 1.25p$.

33 [Intermediate] • [MCQ] • Target: Updating Component Ratios

An alloy consists of copper and zinc in a $7:3$ ratio by mass. If $6\,\text{kg}$ of zinc is added to $40\,\text{kg}$ of this alloy, what is the new ratio of copper to zinc?

a. $7:4$

b. $14:9$ ✓

c. $7:5$

d. $2:1$

Initially, $40\,\text{kg}$ contains $\frac{7}{10}(40) = 28\,\text{kg}$ copper and $\frac{3}{10}(40) = 12\,\text{kg}$ zinc. Adding $6\,\text{kg}$ zinc gives $18\,\text{kg}$ zinc, so the new ratio is $28:18 = 14:9$.

34 [Intermediate] • [MCQ] • Target: Density, Mass & Volume Unit Conversions

Titanium has a density of $4.5\,\text{g/cm}^3$. What is the mass, in kilograms, of a solid titanium block with volume $2{,}000\,\text{cm}^3$?

a. $0.9\,\text{kg}$

b. $4.5\,\text{kg}$

c. $9.0\,\text{kg}$ ✓

d. $90\,\text{kg}$

Mass $= 4.5\,\text{g/cm}^3 \times 2000\,\text{cm}^3 = 9{,}000\,\text{g} = 9.0\,\text{kg}$.

35 [Intermediate] • [MCQ] • Target: Opposing Rates (Fill vs. Drain)

Pipe A can fill a tank in 4 hours, while a drain valve B can empty the full tank in 6 hours. If both Pipe A and drain valve B are accidentally left open when the tank is empty, how many hours will it take to fill the tank?

a. 8

b. 10

c. 12 ✓

d. 24

Net filling rate $= \frac{1}{4} - \frac{1}{6} = \frac{3-2}{12} = \frac{1}{12}$ tank per hour. Therefore it takes 12 hours to fill the tank.

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36 [Intermediate] • [MCQ] • Target: Chained Percentage Multipliers

If a is 25% greater than b, and b is 20% less than c, then a is what percent of c?

a. 95% **b.** 100% ✓

c. 105% **d.** 110%

Chain the multipliers: $a = 1.25b = 1.25(0.80c) = 1.00c$, so a is 100% of c.

37 [Intermediate] • [MCQ] • Target: Harmonic Mean for Equal-Distance Round Trips

A cyclist rides uphill at 12 km/h and immediately rides back down the exact same trail at 36 km/h. What is the cyclist's average speed for the entire round trip?

a. 18 km/h ✓ **b.** 20 km/h

c. 24 km/h **d.** 27 km/h

For equal distances at speeds v_1 and v_2, the average speed is the harmonic mean: $\frac{2v_1v_2}{v_1+v_2} = \frac{2(12)(36)}{12+36} = \frac{864}{48} = 18$ km/h.

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Multivariable Scaling of 3D Cylinder Volume • Cross-Lesson: Topic 8.01 (Volume of 3D Solids)

If the radius of a right circular cylinder is increased by 20% and its height is decreased by 25%, by what percentage does the volume of the cylinder change?

a. Decreases by 5% **b.** **Increases by** 8% ✓

c. Increases by 10% **d.** Decreases by 10%

Since $V = \pi r^2 h$, the volume multiplier is $(1.20)^2(0.75) = 1.44 \times 0.75 = 1.08$, which is an 8% increase.

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39 [Challenging] • [MCQ] • Target: Drain-and-Replace Mixture Modeling • Cross-Lesson: Topic 1.01 (Linear Conservation Equations)

A 10-liter radiator is filled with a 60% antifreeze solution. How many liters of the solution must be drained and replaced with pure (100%) antifreeze so that the resulting 10-liter mixture is 75% antifreeze?

a. 2.5 L

b. 3.25 L

c. 3.75 L ✓

d. 4.0 L

If x liters are drained and replaced with pure antifreeze, the antifreeze balance is

$0.60(10 - x) + 1.00x = 0.75(10) \Rightarrow 6 + 0.40x = 7.5 \Rightarrow x = 3.75\,\text{L}$.

40 [Challenging] • [MCQ] • Target: Compound Percentage Extrapolation • Cross-Lesson: Topic 4.01 (Exponential Growth)

An investment grows at a constant annual compound rate so that its value increases by 44% over 2 years. What is the percentage increase over 3 years?

a. 66%

b. 72.8% ✓

c. 88%

d. 95.2%

Since $(1 + r)^2 = 1.44$ with $r > 0$, we have $1 + r = 1.20$ (20% per year). Over 3 years, $(1.20)^3 = 1.728$, which is a 72.8% increase.

41 [Challenging] • [MCQ] • Target: Area Ratios in Similar Triangles • Cross-Lesson: Topic 7.01 (Similar Triangles)

In ΔABC, point D lies on AB such that $AD : DB = 2 : 3$, and point E lies on AC such that $DE \parallel BC$. The area of ΔADE is what percent of the area of trapezoid $DBCE$?

a. 16%

b. 19.05% ✓

c. 40%

d. 66.7%

Since $AD : AB = 2 : (2 + 3) = 2 : 5$ and $\Delta ADE \sim \Delta ABC$, the area ratio is $[ADE] : [ABC] = 2^2 : 5^2 = 4 : 25$. Thus trapezoid $DBCE$ has relative area $25 - 4 = 21$, and $\frac{4}{21} \times 100\% \approx 19.05\%$.

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42 [Challenging] • [MCQ] • Target: Symmetric Percentage Rise & Fall via Quadratics • Cross-Lesson: Topic 3.01 (Difference of Squares)

A store raises the price of an item by $x\%$ and later reduces the new price by $x\%$, resulting in a final price that is 9% less than the original price. What is the value of x?

a. 9

b. 15

c. 30 ✓

d. 45

The net multiplier is $(1+\frac{x}{100})(1-\frac{x}{100}) = 1-\frac{x^2}{10000} = 1-0.09 \Rightarrow \frac{x^2}{10000} = 0.09 \Rightarrow x^2 = 900 \Rightarrow x = 30$.

43 [Challenging] • [MCQ] • Target: Ratio Rebalancing & Transfer • Cross-Lesson: Topic 2.01 (Linear Systems)

In a survey of 500 voters, the ratio of voters favoring Candidate A to Candidate B is $7:3$. How many voters must switch their support from Candidate A to Candidate B so that the new ratio becomes $1:1$?

a. 50

b. 75

c. 100 ✓

d. 125

Initially, Candidate A has $\frac{7}{10}(500) = 350$ voters and Candidate B has 150. For a $1:1$ split (250 each), $350 - 250 = 100$ voters must switch.

44 [Challenging] • [MCQ] • Target: Gear Ratios & Angular Velocity in Radians • Cross-Lesson: Topic 9.01 (Circles & Radian Measure)

Two gears are meshed together. Gear A has 36 teeth and rotates at 150 rpm. If Gear B has 24 teeth, through how many radians does Gear B rotate in 2 seconds?

a. 10π

b. 15π ✓

c. 20π

d. 30π

By inverse tooth ratio, Gear B rotates at $150 \times \frac{36}{24} = 225$ rpm $= \frac{225}{60} = 3.75$ rev/s. In 2 seconds, Gear B completes 7.5 revolutions, which is $7.5 \times 2\pi = 15\pi$ radians.

45 [Challenging] • [MCQ] • Target: Linear Invariance of Equal Proportions • Cross-Lesson: Topic 10.01 (Rational Expressions)

To satisfy the synthesis condition, if $\frac{a}{b} = \frac{c}{d} = \frac{3}{4}$ with $b + 2d \neq 0$, what is the value of $\frac{a+2c}{b+2d}$?

a. $\frac{3}{8}$

b. $\frac{3}{4}$ ✓

c. $\frac{9}{16}$

d. $\frac{6}{7}$

Since $a = \frac{3}{4}b$ and $c = \frac{3}{4}d$, we have $\frac{a+2c}{b+2d} = \frac{\frac{3}{4}(b+2d)}{b+2d} = \frac{3}{4}$ by the mediant property of equal ratios.

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